Vertex cover Approximation using Linear Programming

- Given an undirected graph G = (V, E), the vertex cover problem asks for the subset $S \subseteq V$, s.t., all the edges have at least one vertex in S.
- We want to minimize the number of elements in S.
- In the weighted version, there is a weight given on every vertex (w_v, $\forall v \in V$) and the task is to minimize the weight of S, $w(S) = \Sigma_{v \in S} w_v$.
- We can without loss of generality assume that $\Sigma_v w_v = 1$.
- Clearly this is an optimization problem.
- It is known to be NP-hard and so not expected to have a polynomial time algorithm.
- We are interested in finding an approximation algorithm for this problem.

Integer programming formulation

- An integer program is an optimization problem where variables are constrained to be a set of integers.
- The vertex cover problem can be framed as an integer programming problem.
- Given an undirected graph G = (V, E), which has |V | = n vertices and |E| = m edges, introduce variable y_v for every vertex $v \in V$. Say $y_v = 1$ if the vertex is in S and $y_v = 0$ otherwise.
- Suppose w_v is the weight of the vertex v, then the task is to

minimize the weight of elements in S.

- The constraint is that every edge should have at least one vertex in S.
- This can be expressed by saying, $\forall \{u, v\} \in E, y_u + y_v \geq 1$.
- So the integer program is,

$$\min \Sigma_{v \in V} w_v y_v$$

$$\text{s.t. } y_u + y_v \geq 1 \quad \forall \{u, v\} \in E$$

$$y_v \in \{0, 1\} \quad \forall v \in V \qquad (1)$$

LP relaxation and rounding

- How can we convert this integer program into an LP.
- The relaxation of an optimization program is another optimization program which ideally is easier to solve and every solution of original program is also a solution of the new program with the same (or related) objective value.
- In the case of Eqn. 1, we change the domain of y_i's to be [0, 1] instead of integers {0, 1}.

$$\min \Sigma_{v \in V} w_v y_v$$

$$\text{s.t. } y_u + y_v \geq 1 \quad \forall \{u, v\} \in E \qquad (2)$$

$$y_v \in [0, 1] \quad \forall v \in V$$

- The only change is that the y_i's are real numbers between 0 and 1 instead of integers now.
- It is clear that any solution of Eqn. 1 will still be a solution of Eqn. 2 with same objective value.

- Hence the objective value of the LP is at least the value of the integer program.
- The LP is called the relaxation of the integer program.
- Since the LP can be solved in polynomial time (with some precision), we can obtain the optimal solution of Eqn. 2.
- The goal now is to convert these fractional solutions into integer ({0, 1}) solutions.
- The process of converting fractional solutions into integers is called rounding.
- We now use the following algorithm:

Vertex Cover via LP:

Solve LP to obtain an optimal fractional solution y^*

Let $S = \{v \mid y^*_v \geq 1/2\}$

Output S

- Suppose the solution of LP gives y^* as the optimal solution.
- One simple rounding procedure could be, if $y^*_v \geq 1/2$ then we include the vertex in the vertex cover otherwise skip it.
- The corresponding integer solution, say z, is

$$z_v = \begin{cases} 1 \text{ if } y^*_v \geq 1/2 \\ 0 \text{ if } y^*_v < 1/2 \end{cases}$$

- It is easy to show the following two properties.
  - z is a feasible solution.
  - The objective value of z, call it $Ob(z) = \Sigma_v w_v z_v$, is at most

twice the objective value of Ob(y*).

- This gives us a 2-approximate algorithm for vertex cover.
- $Ob(z) \leq 2\, Ob(y^*) = 2\, Opt(LP) \leq 2\, Opt(IP)$
- Here Opt(LP), Opt(IP) denote the optimal values of the linear program (Eqn. 2) and the integer program(Eqn.1)